WORKED SOLUTION

STEP I Specimen paper, Question 5

Pure • Counting Digits and Factorials

STEP I Specimen paper, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q5

The key idea

Both parts are about counting carefully rather than computing. In (i), first find exactly where the 1000th digit falls by counting digits in blocks (one-digit, two-digit, three-digit numbers), then count the 7s place by place: units, tens, hundreds. In (ii), each trailing zero is a factor of $10 = 2 \times 5$, and fives are scarcer than twos, so we just count the factors of 5 in 365!.

Solution

Part (i)

Where do we stop? The numbers 1 to 9 use 9 digits. The 90 two-digit numbers 10 to 99 use 180 digits, giving 189 so far. That leaves

$$1000 - 189 = 811 = 3 \times 270 + 1$$

digits. The first 270 three-digit numbers are $100, 101, \ldots, 369$, and they use 810 digits. The last digit is the first digit of 370, which is a 3. So we have written all of $1, 2, \ldots, 369$ in full, followed by a single 3.

? **Ask yourself** The number 370 has a 7 in it. Does it count?

No. We stop after writing its first digit, the 3, so its 7 is never written. This is exactly why we need to pin down the stopping point to the digit.

Counting the 7s in 1 to 369. We count by place value. Every 7 is in exactly one place, so nothing is counted twice; a number like 77 correctly contributes two.

- *Units digit.* In each complete block of ten, $10k, \ldots, 10k + 9$, exactly one number ends in 7. The numbers 0 to 359 form 36 complete blocks, giving 36 sevens: $7, 17, \ldots, 357$. The partial block 360 to 369 adds one more, 367. Total: 37.
- *Tens digit.* In each complete hundred, exactly ten numbers have tens digit 7: the

numbers 70–79, 170–179 and 270–279. The block 370–379 is beyond 369. Total: 30.

- *Hundreds digit.* Every number up to 369 has hundreds digit at most 3 (or no hundreds digit at all). Total: 0.

The final lone 3 adds nothing. Hence the digit 7 is used

$$37 + 30 + 0 = 67 \text{ times.}$$

Part (ii)

The number of trailing zeros of N in base 10 is the largest k with $10^k \mid N$. Since $10 = 2 \times 5$, this is the smaller of the power of 2 and the power of 5 in the prime factorisation of N.

Counting the fives. The power of 5 dividing 365! is the sum, over $j = 1, \ldots, 365$, of the number of factors of 5 in j. A multiple of 5 contributes at least one factor, a multiple of 25 contributes a second, a multiple of 125 a third, and so on. Counting each layer separately:

$$\begin{aligned}
\left\lfloor \tfrac{365}{5} \right\rfloor &= 73 && \text{(multiples of 5),} \\
\left\lfloor \tfrac{365}{25} \right\rfloor &= 14 && \text{(multiples of 25),} \\
\left\lfloor \tfrac{365}{125} \right\rfloor &= 2 && \text{(multiples of 125: 125, 250),} \\
\left\lfloor \tfrac{365}{625} \right\rfloor &= 0.
\end{aligned}$$

A number divisible by exactly 5^m is counted once in each of the first m lines, so it contributes exactly m, as it should. The power of 5 in 365! is therefore

$$73 + 14 + 2 = 89.$$

The twos are not the bottleneck. The same argument for 2 gives at least $\lfloor 365/2 \rfloor = 182 > 89$ factors of 2. (In fact there are $182 + 91 + 45 + 22 + 11 + 5 + 2 + 1 = 359$.) So $2^{89} \cdot 5^{89} = 10^{89}$ divides 365!, but 10^{90} does not, since $5^{90} \nmid 365!$.

Hence 365! ends in exactly **89** zeros.

Checks

For (i), a quick check on the stopping point: $9 + 180 + 3 \times 270 = 999$ digits take us to the end of 369, and the next digit is the leading 3 of 370. Writing out the first 1000 digits by computer and counting confirms 67 sevens, with 20 of them in 1–99 and 47 in 100–369. For (ii), compute 365! exactly (it has 779 digits) and count its trailing zeros. There are 89, matching the count of fives. A small case to test the method by hand: 25! should end in $\lfloor 25/5 \rfloor + \lfloor 25/25 \rfloor = 6$ zeros, and indeed $25! = 15\,511\,210\,043\,330\,985\,984\,000\,000$.

Common slips

- Counting 370 (or even 371–379) as written, which gives 68 or more.

- Using only $\lfloor 365/5 \rfloor = 73$ and forgetting that 25, 50, … contain two fives and 125, 250 contain three.
- Counting 7s by numbers instead of by digits, so that 77, 177 and 277 are counted once rather than twice.

Beyond STEP

The layered count in (ii) is Legendre's formula: the power of a prime p in $n!$ is $\sum_{k \geq 1} \lfloor n/p^k \rfloor$, which also equals $(n - s_p(n))/(p-1)$, where $s_p(n)$ is the sum of the base-p digits of n. Here $365 = 2430_5$, with digit sum 9, and $(365 - 9)/4 = 89$. The formula underlies Kummer's theorem on the prime factors of binomial coefficients. Davenport's *The Higher Arithmetic* is a lovely place to explore this kind of elementary number theory further.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.